Group 6

Smart Bottle Bank and Waste-to-Energy Gasification

Final project report for mentors

October 2026

1. Summary

The project has two connected parts. Part A is a solar-powered smart bottle bank that collects plastic bottles, counts them, measures how full it is and reports its battery and charging state to an admin dashboard. Part B is a 1 kW demonstration waste-to-energy unit that turns dry biomass pellets into electricity by controlled downdraft gasification. The collected bottles are not used in Part B; its fuel is dry biomass, paper or cardboard pellets made from other campus waste.

Deliverable	Status
Bottle bank design (CAD macro, 3D model, electronic schematic)	Designed. CAD macro logic-checked only; not yet run in FreeCAD by us
Admin dashboard (7 campus locations)	Working on simulated data; real data needs firmware and a server
Bottle bank firmware (counting, telemetry)	Earlier version written; update for the new design still to do
Gasifier sizing, process flow diagram, CAD macro, 3D model	Preliminary design. Sizes come from published rules and must be confirmed by tests
Prototypes	Not built yet

2. Part A: Smart bottle bank

2.1 What it does

Bottles are dropped through a wide inlet high on the front and slide down an open chute into the bin. There is no reward system and no one-bottle-at-a-time gate. An infrared beam across the chute counts every bottle, and an ultrasonic sensor above the bin measures how full it is. A small controller (ESP32) sends the count, fill level, battery state, door state and signal strength to the admin dashboard.

2.2 Mechanical design



Item	Value
Body	Closed cylinder, 650 mm diameter, 1550 mm tall, on 4 legs
Inlet	160 mm diameter, about 1.27 m above the floor
Bin	About 900 mm tall, about 270 litres
Capacity	About 210 half-litre bottles (loose packing); fewer 1.5 L bottles
Top	Solar panel, 30 W, tilted 10 degrees
Front	Label, 20x4 LCD, green and red LEDs
Access	Rear service door with a door switch
Inside	Angled chute, IR counting beam, ultrasonic sensor, electronics box, 12 V battery

Figure 1. Bottle bank (3D render)

The design is a FreeCAD macro (BottleBank_FreeCAD.py) that builds every part and exports a STEP file, which Onshape can import. An interactive 3D page shows the design with an X-ray view of the inside.

2.3 Electronics

Electronic schematic (v3: collection and telemetry)

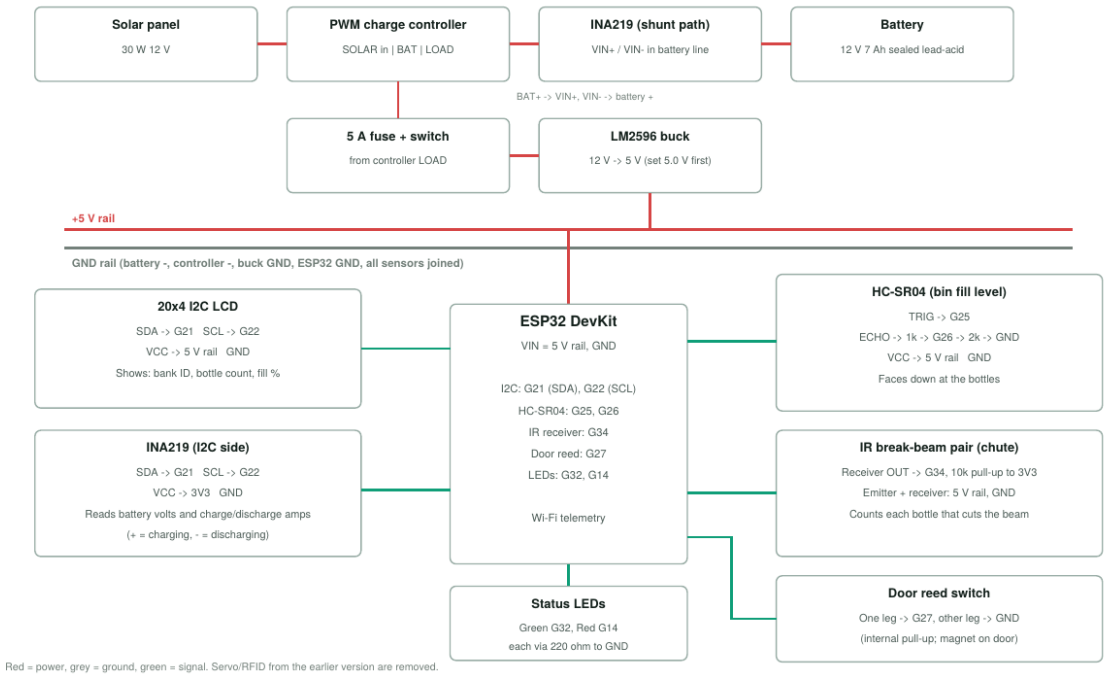


Figure 2. Bottle bank electronic schematic

The 30 W solar panel charges a 12 V 7 Ah battery through a PWM charge controller. An INA219 sensor in the battery line measures battery voltage and whether the battery is charging or discharging (above +50 mA = charging). A buck converter supplies 5 V. Bottle count comes from

the IR beam, fill level from the HC-SR04 ultrasonic sensor, and door-open status from a reed switch. The IR receiver output needs a 10 k pull-up to 3.3 V (not 5 V).

2.4 Admin dashboard

One web page shows all banks: total bottles, banks online, average fill level, pickups needed, and low-battery banks. A map and a table list each bank with its location, bottle count, fill level, battery percentage and voltage, charging status, signal, last-seen time and status. Alerts are raised when a bank is offline, nearly full, has its door open or has a low battery. The seven locations are Engineering Lecture Theatre (LTH), Faculty of Agriculture, Environmental Science, ECE Classroom, CHE Classroom, Female Hostel and Middle Block.

Note: all numbers are simulated, and the map positions are placeholders until real GPS coordinates are measured at each location.

2.5 Power estimate

The controller, LCD and sensors draw about 2 W (about 48 Wh a day). A 30 W panel in Lagos (about 4.5 peak-sun hours, 70% system efficiency) gives about 95 Wh a day. On rainy days the panel gives much less, and the 84 Wh battery lasts about two days, so the LCD backlight should switch off when idle. These are estimates to be confirmed by measurement.

3. Part B: Waste to energy

3.1 Waste to pellets

1. Collect and sort dry paper, cardboard, sawdust, wood scraps and farm residues (for example groundnut shells or corn cobs). Remove plastics, metal, food waste and anything wet.
2. Dry to about 10-12% moisture in the sun or a warm dryer.
3. Shred and grind to a few millimetres.
4. Add a little cassava starch as binder (a few percent) for paper and farm residues; wood and sawdust bind on their own.
5. Press into pellets of about 7 mm with a small pellet mill or press; cool, sieve out dust, and store dry.

The throat of the gasifier is only 3.5 cm, so pellets must be about 7 mm, hard, and under 15% moisture. A 4-hour run needs about 9 kg of pellets.

3.2 How the gasifier works

Pellets enter the hopper and dry. Lower down they break down (pyrolysis) and burn partly where air enters through three nozzles. The hot gas then passes over glowing charcoal and becomes syngas (carbon monoxide, hydrogen and methane). The gas is cleaned and cooled in a cyclone, gas cooler, water scrubber and fine filter, stored in a small buffer, and burned in a petrol generator that has been set up to run on the gas. A flare burns the gas during start-up and shut-down.

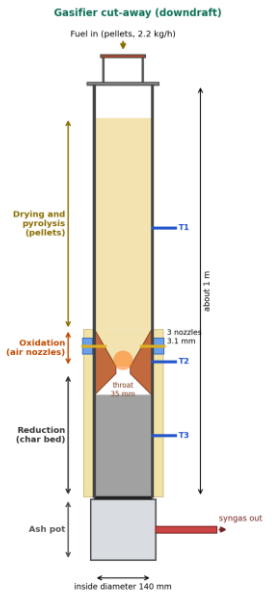
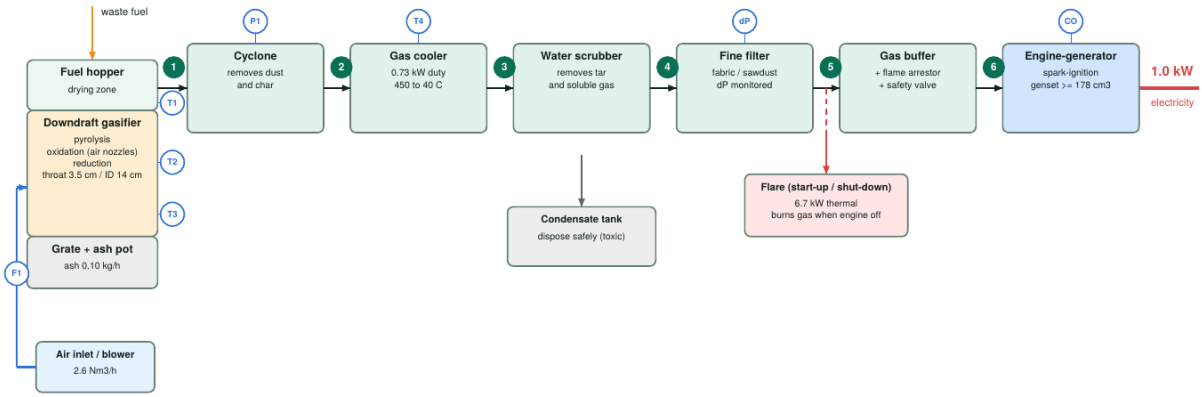


Figure 3. Gasifier cut-away

Design basis (1 kW unit)	Value
Electrical output	1 kW
Fuel	2.2 kg/h as fed (1.96 dry); about 8.9 kg per 4-hour run
Syngas	5.3 Nm3/h (6.7 kW thermal)
Air	2.6 Nm3/h through 3 nozzles of 3.1 mm
Reactor	140 mm inside diameter, about 1.05 m tall, hopper 9.3 L
Throat	35 mm; hearth load 0.55 Nm3/cm2/h (accepted range 0.3-0.9)
Specific gasification rate	127 kg/h/m2 (accepted range 110-210)
Engine	Genset of at least 178 cm3 (typically 2.5-3 kVA petrol)
Efficiency	10.5% fuel to electricity (typical for small units)
By-products	Ash 0.1 kg/h; cooler duty 0.7 kW; tar water



Blue tags = instruments (T temperature, P pressure, dP pressure drop, F air flow, CO carbon monoxide alarm). Red = gas to flare / electricity. Green numbers = streams (table below).

#	Stream	Amount	Conditions
1	Waste fuel (dry biomass, paper or cardboard pellets)	2.23 kg/h as fed (1.96 kg/h dry)	12% moisture, pellets 6-10 mm
2	Gasification air	2.6 Nm3/h (3.4 kg/h)	ER 0.3, through 3 nozzles of 3.1 mm
3	Hot raw syngas	5.3 Nm3/h, 6.7 kW thermal	about 450 C, carries dust and tar
4	Cooled gas	5.3 Nm3/h	about 40 C after cooler
5	Clean gas	5.3 Nm3/h	after scrubber and filter
6	Engine intake (gas + air)	11.2 Nm3/h	about 1.1 : 1 air to gas

Figure 4. Process flow diagram with instruments (T temperature, P pressure, dP filter pressure drop, F air flow, CO alarm)

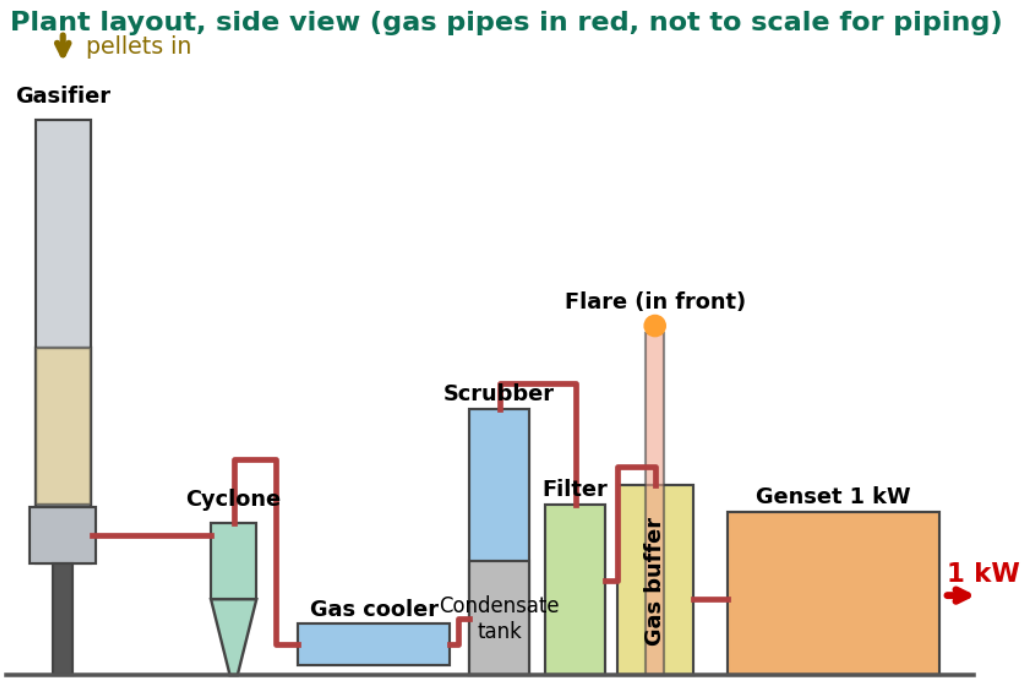


Figure 5. Plant layout, side view

The plant is also available as a FreeCAD macro (Gasifier_FreeCAD.py, STEP export) and as an interactive 3D page. Sizes come from a spreadsheet (Gasifier_Sizing.xlsx) where every assumption can be changed and the checks update. The design rules are from FAO Forestry Paper 72 and published papers for larger units, so they are a starting point.

3.3 Safety

- Syngas contains about 20% carbon monoxide, which is poisonous, odourless and explosive.
- Operate outdoors only, with CO alarms, a flare, a flame arrestor and a trained supervisor present. Leak-test every joint and purge before starting.
- Only dry biomass, paper or cardboard pellets are used. Plastic bottles are never fed: they melt, block the bed and give toxic gases.
- Hot surfaces, ash and tar water need guards, cooling time and proper disposal.

4. How the two parts fit together

Both parts reduce waste on campus. The bottle bank keeps plastic bottles out of the waste stream for recycling, and the gasifier turns other dry waste into electricity. They are separate processes. A natural next step is to add the gasifier sensors (temperatures, gas alarm, air flow) to the same admin dashboard.

5. Limitations and next steps

Item	Next step
Bottle bank firmware for the new design (counting, fill, telemetry upload)	Write and test, first in the Wokwi simulator, then on a prototype
Server for real telemetry	Choose a host and point the dashboard to it
Real GPS coordinates	Measure each location on campus
Wi-Fi coverage at all seven locations	Test; use a GSM module where there is none
FreeCAD models	Run in FreeCAD, check the fit of every part, then fabricate

Item	Next step
Gasifier sizing	Confirm by test runs; measure fuel use, gas quality and tar
Gasifier electronics and dashboard integration	Design thermocouple, pressure, flow and CO monitoring
Risk assessment	Review by a qualified supervisor before any hot test

6. Files and links

Bottle bank 3D design:
https://drive.google.com/file/d/1SzP1PL08sMHCqb8LFmC0j5EfFwXUmCj1/view?usp=drive_link

Admin dashboard: https://fash0404.github.io/Bottle-bank-dashboard/

Gasification plant 3D design https://drive.google.com/file/d/1GrQGRY7Cd4wNcZdrwrQuniDUI-qUKGne/view?usp=drive_link